# Zen Nguyen

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## TECHNICAL SKILLS

- **Languages:** Python, Java, C++, SQL
- **Backend & Systems:** gRPC, RESTful APIs
- **Infrastructure & DevOps:** Docker, AWS
- **Testing & Validation:** Test plans, Test reports, Validation

## TECHNICAL PROJECTS

**TitanSwarm (Autonomous AI Co-Pilot)** Jan 2026 – Present

*Personal Project · Python, LangChain, Streamlit, Gemini API*

- Automated end-to-end job application workflows by architecting an autonomous AI Co-Pilot, enabling the discovery, extraction, and parsing of real-time Software Engineering postings.
- Developed a zero-hallucination RAG engine using LangChain, Gemini APIs, and FAISS Vector Store, delivering uniquely tailored, ATS-optimized PDF resumes for each job using only verified user data.
- Designed and implemented a Streamlit Human-in-the-Loop dispatch UI with a Kanban board, enabling users to review, tailor, and manage applications across a full pipeline.
- Applied Test-Driven Development (TDD) with a 113-test suite using Pytest to validate all components including the scraper, AI tailor, PDF generator, and repository layer, ensuring robust **validation**.

**TitanStore (Distributed Raft Database)** Jan 2026 – Present

*Personal Project · Go, Raft Consensus Algorithm, TCP Sockets, WAL*

- Developed a distributed key-value database in Go from scratch, implementing the **Raft consensus protocol** for leader election and fault-tolerant log replication across a cluster of nodes.
- Engineered a crash-safe Write-Ahead Log (WAL) with atomic snapshots to guarantee data durability and enable node recovery after failures without data loss.

**Gridlock Casino (2D Arcade Engine)** Feb 2026 – Present

*Collaborative Project · Java, JUnit, Maven, BFS*

- Architected a custom 2D grid-based arcade game engine in Java, managing game state, 60-FPS rendering cycles, and concurrent user input within a strict MVC architecture.
- Implemented core gameplay algorithms, including BFS (Breadth-First Search) for autonomous enemy pathfinding, hitscan vector math for projectile collision, and dynamic grid masking for fog-of-war visibility.

## EDUCATION

**Bachelor of Science, Computing Science** May 2025 – Present

*Simon Fraser University*

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 – Apr 2025